The center of mass is a k-dimensional point, each of whose coordinates is the mean of the coordinates of all the other children's centroids[10].

Once we have the new centroid, we need to update the radius of the node's bounding envelope: this is defined as the minimum radius for which the bounding envelope includes all the bounding envelopes for current node's children; in turn, we can define it as the maximum distance between current node's centroid and any point in its children. Figure 10.17 shows why these distances are, for each child, the sum of the distance between the two centroids, plus the child's radius (as long as we assume that points have radius==0, this definition also works for leaves).

10.3.4 Insertion: Split Nodes

We can now move to the implementation of method `split`, in listing 10.11.

```
Listing 10.11 The SsNode::split method
function
SsNode::split()
      #1
  splitIndex ← this. findSplitIndex(coordinateIndex)
                          #2
  if this.leaf then
    newNode1 ← new SsNode(true,
points=this.points[0..splitIndex-1])          #3
    newNode2 ← new SsNode(true,
points=this.points[splitIndex..])             #4
  else
    newNode1 ← new SsNode(false, children=this.children[0..
index-1])          #5
```